FitBuddy: AI Fitness Plan Generator using Gemini Models
Government Arts and Science College , Manur - MSU142
Team Id:6ab21893a614981d6d2c53e1
Team Name:7
Team Leader:Mohamed Mujahith Kasim M, NMID:DD48BB73EA4A64C8B904D2400C9F7D6F
Member 2:Suman M ,NM ID: 0192097D4D7A60988DA49766C01835E0
Member 3:Udhayanithi S ,NMID:52361B3C0E57DDDB4F242E43149A5A47
Member 4:Malai raja A ,NMID:7255103A926613882F2472E0E276D163

Project Report – FitBuddy: AI Fitness Plan Generator using Gemini Models

1. Introduction

FitBuddy – AI Fitness Plan Generator using Gemini Models is a web-based application designed to generate personalized fitness plans using Artificial Intelligence. The application uses user information such as **age, weight, fitness goal, and preferred workout intensity** to generate a customized 7-day workout plan. It also provides nutrition or recovery tips according to the user's selected goal.

2. Problem Statement

Many users find it difficult to create a structured workout routine that matches their individual fitness goals and preferred intensity. FitBuddy addresses this requirement by using generative AI to create personalized workout schedules and allowing users to modify their plans through feedback.

3. Objectives

The main objectives of FitBuddy are:

- To generate personalized 7-day workout plans.
- To provide goal-based nutrition and recovery tips.
- To allow users to modify their workout plans using feedback.
- To store user details and generated plans.
- To provide an administrative interface for viewing users and their plans.
- To create a simple and user-friendly fitness planning application.

4. Technologies Used

Component	Technology
Programming Language	Python
Backend	FastAPI
AI Models	Google Gemini
Frontend	HTML, CSS, Jinja2
Database	SQLite
ORM	SQLAlchemy
Server	Uvicorn
Version Control	Git

These technologies are identified in the project's prerequisites and architecture.

5. System Architecture

FitBuddy follows a modular architecture consisting of:

Frontend → FastAPI Backend → Gemini AI → SQLite Database

- **Frontend:** Collects user information and displays results.
- **FastAPI:** Handles routing, input processing, and application logic.
- **Gemini AI:** Generates workout plans and nutrition tips.
- **SQLite/SQLAlchemy:** Stores user information and generated plans.

6. Project Development

Workout Generation

The `generate_workout_gemini()` function uses Gemini 1.5 Pro to generate a structured 7-day workout plan. The plan includes warm-up, main workout exercises, sets/repetitions, and cooldown or recovery guidance.

Nutrition Tip Generation

The `generate_nutrition_tip_with_flash()` function uses Gemini Flash to provide concise nutrition or recovery suggestions based on the user's fitness goal.

Feedback-Based Updating

Users can submit feedback such as adding yoga or increasing cardio. The system sends the original plan and feedback to Gemini 1.5 Pro and generates an updated workout plan.

Database

User information and AI-generated plans are stored using SQLite and SQLAlchemy. The system preserves both original and updated workout plans.

7. Major Modules

1. **User Input Module**
2. **Workout Generation Module**
3. **Nutrition/Recovery Module**
4. **Feedback Module**
5. **Database Module**
6. **Admin Module**
7. **Frontend Interface Module**

8. User Interface

The project contains three major templates:

- **index.html** – Collects user information.
- **result.html** – Displays the generated workout plan, nutrition tip, and feedback form.
- **all_users.html** – Provides the administrator's view of users and their plans.

The interface uses a fitness-oriented design with responsive CSS, centered layouts, input fields, buttons, and result sections.

9. Project Workflow

User Details
↓
FastAPI Backend
↓
Gemini AI Processing
↓
Personalized 7-Day Workout Plan

↓
Nutrition/Recovery Tip
↓
Database Storage
↓
User Feedback
↓
Updated Workout Plan
↓
Admin Dashboard

10. Testing

The application is tested by verifying user input, workout-plan generation, nutrition tips, feedback-based plan updates, database storage, frontend pages, and FastAPI routes. The application can be run locally using Uvicorn, and the FastAPI `/docs` interface can be used to test the API endpoints.

11. Project Demonstration

During demonstration, the following workflow can be shown:

1. Open the FitBuddy home page.
2. Enter user details.
3. Generate the fitness plan.
4. Display the 7-day workout schedule.
5. Display the nutrition/recovery tip.
6. Submit feedback.
7. Generate the updated plan.
8. Open the admin page and display stored user information and plans.

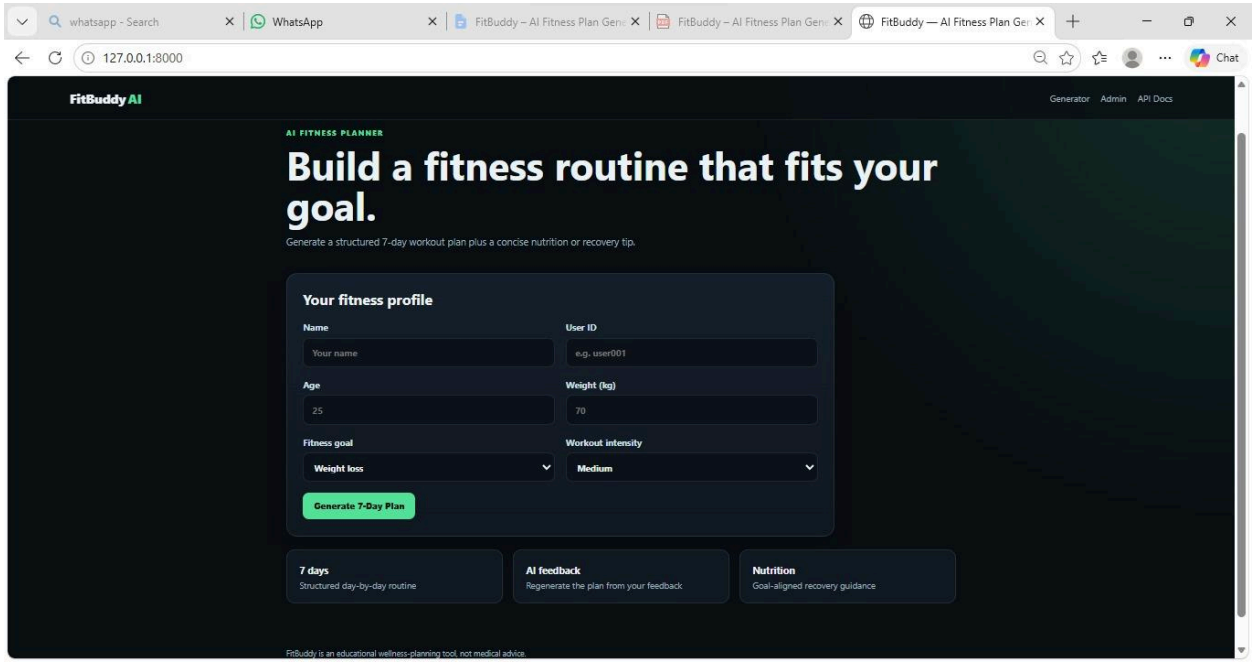
12. Conclusion

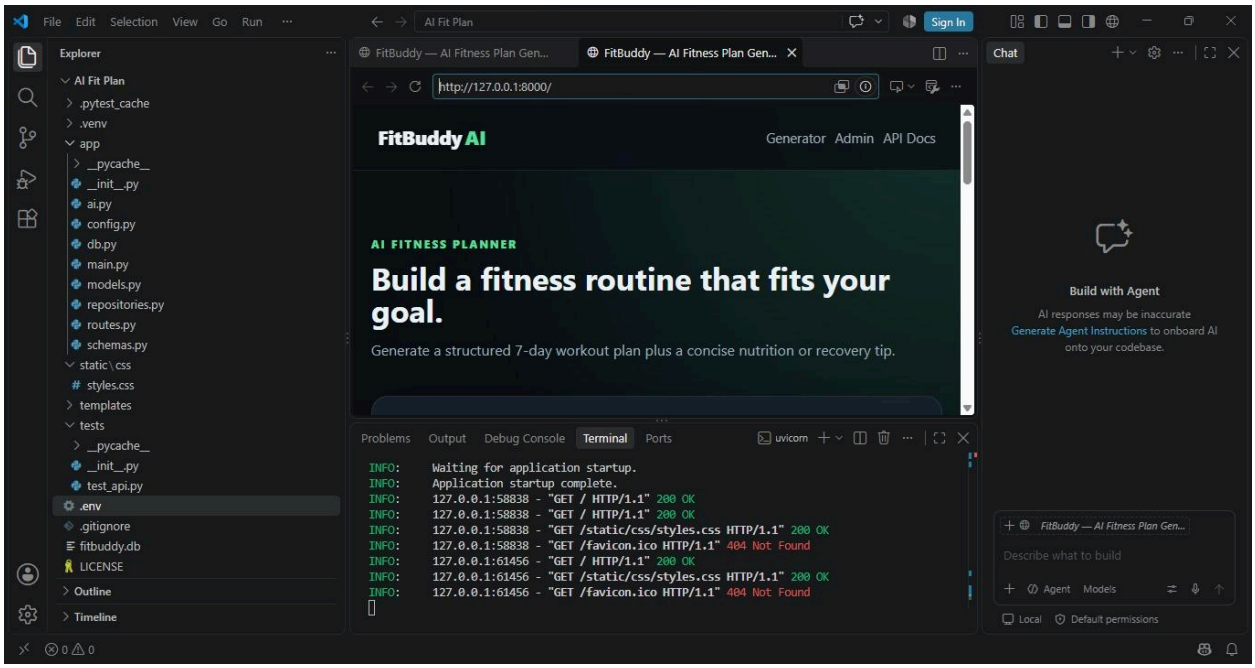
FitBuddy is a comprehensive AI-powered fitness planning web application that combines **Google Gemini models, FastAPI, Jinja2, SQLite, and SQLAlchemy**. It generates personalized 7-day workout plans and nutrition/recovery tips based on user information. Its feedback feature allows workout plans to be updated according to changing user requirements, while the administrative view provides access to stored user and plan information.

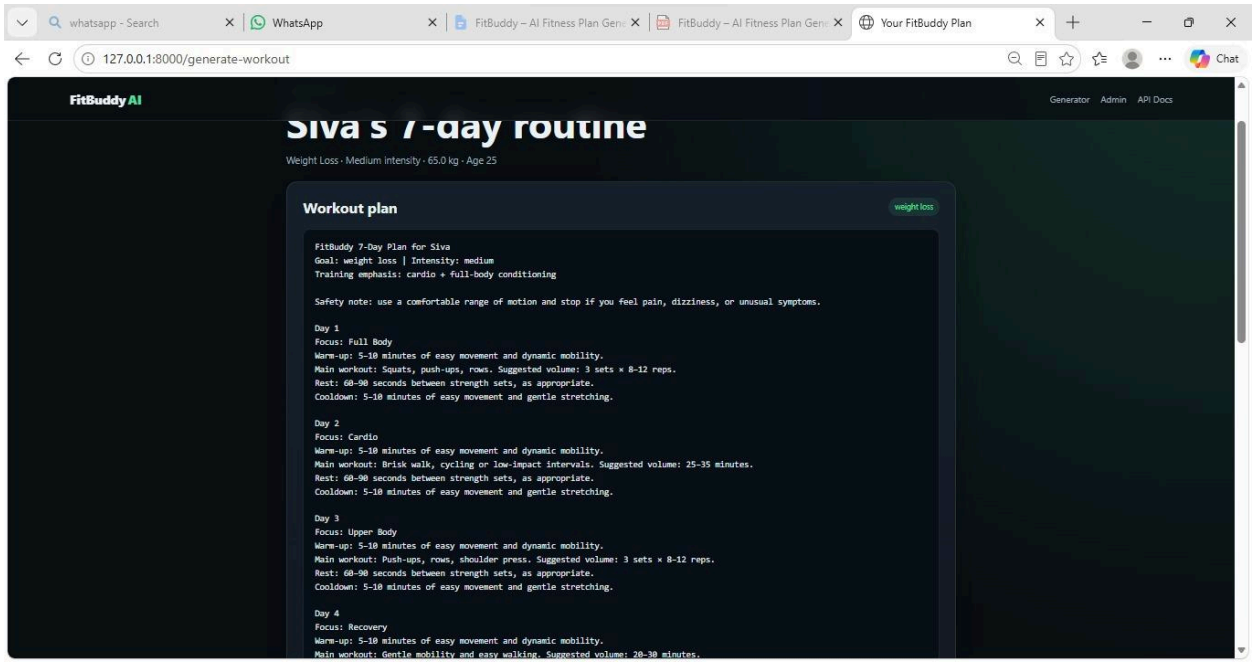
Overall, the project demonstrates how generative AI can be integrated into a web application to provide a personalized and adaptable fitness-planning experience.

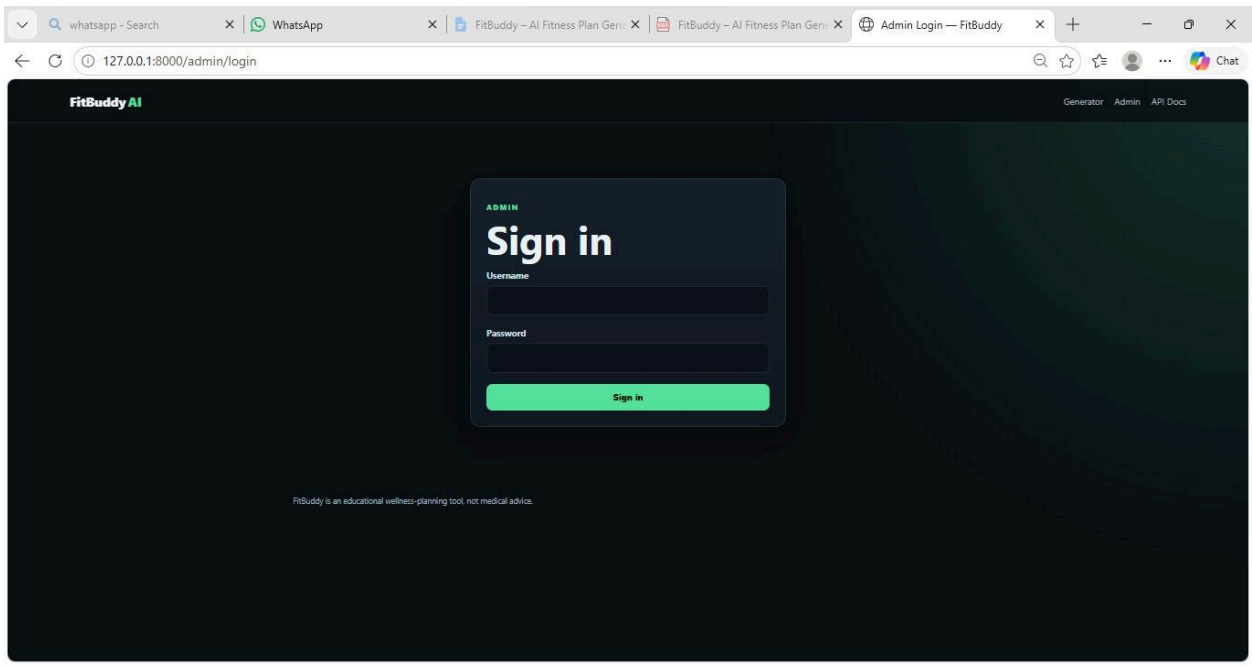
Continue the FitBuddy report

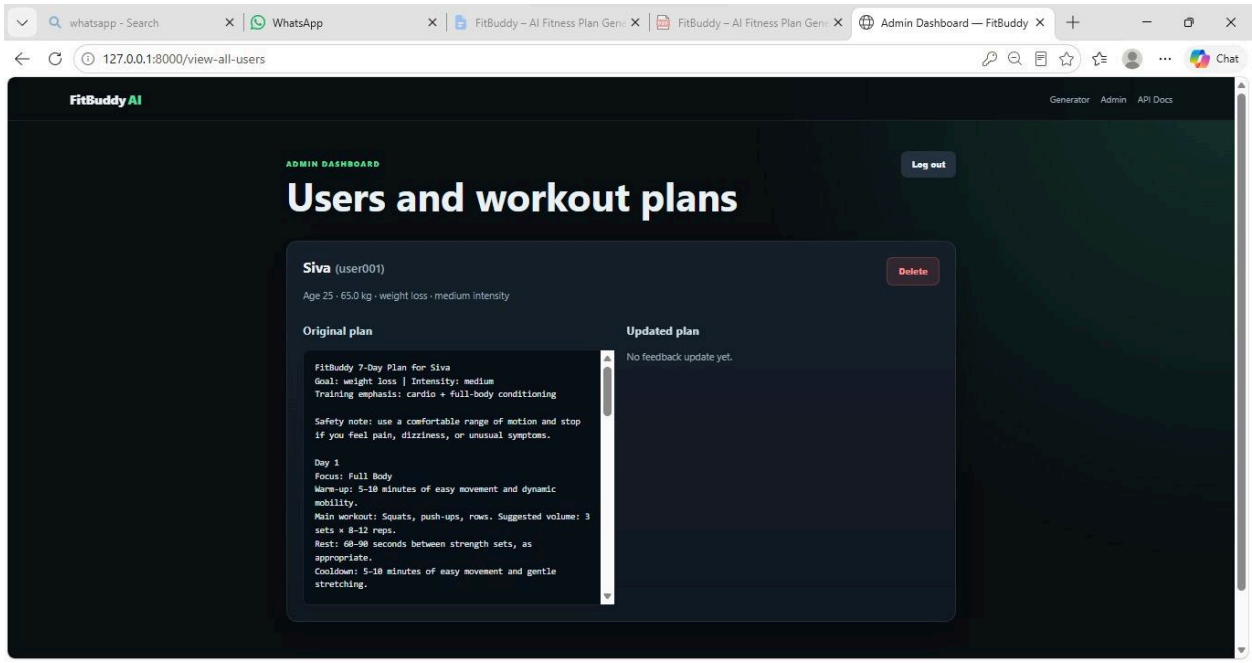
- Create an abstract and keywords
- Add a references section

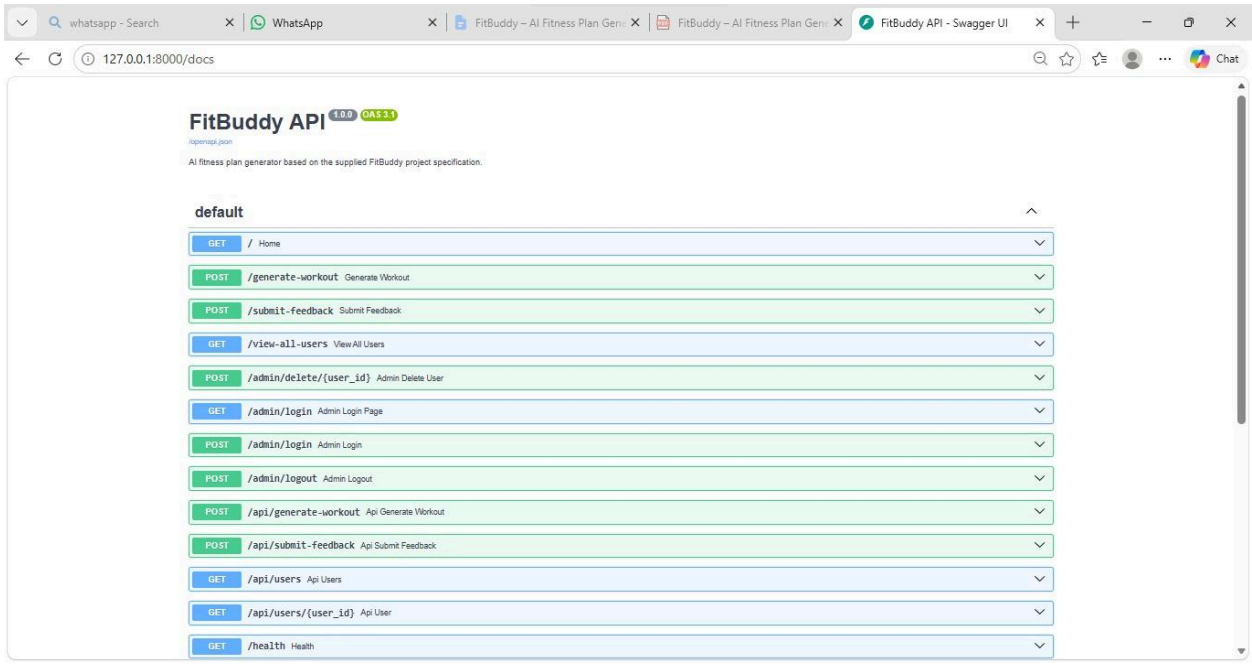





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WhatsApp
FitBuddy – AI Fitness Plan Gene
FitBuddy – AI Fitness Plan Gene
Your FitBuddy Plan
127.0.0.1:8000/generate-workout
Chat
FitBuddy AI
Generator
Admin
API Docs
Siva's 7-day routine
Weight Loss · Medium intensity · 65.0 kg · Age 25
Workout plan
weight loss
FitBuddy 7-Day Plan for Siva
Goal: weight loss | Intensity: medium
Training emphasis: cardio + full-body conditioning
Safety note: use a comfortable range of motion and stop if you feel pain, dizziness, or unusual symptoms.
Day 1
Focus: Full Body
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Squats, push-ups, rows. Suggested volume: 3 sets × 8–12 reps.
Rest: 60–90 seconds between strength sets, as appropriate.
Cooldown: 5–10 minutes of easy movement and gentle stretching.
Day 2
Focus: Cardio
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Brisk walk, cycling or low-impact intervals. Suggested volume: 25–35 minutes.
Rest: 60–90 seconds between strength sets, as appropriate.
Cooldown: 5–10 minutes of easy movement and gentle stretching.
Day 3
Focus: Upper Body
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Push-ups, rows, shoulder press. Suggested volume: 3 sets × 8–12 reps.
Rest: 60–90 seconds between strength sets, as appropriate.
Cooldown: 5–10 minutes of easy movement and gentle stretching.
Day 4
Focus: Recovery
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Gentle mobility and easy walking. Suggested volume: 20–30 minutes.


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WhatsApp
FitBuddy – AI Fitness Plan Gene
FitBuddy – AI Fitness Plan Gene
Admin Login — FitBuddy
127.0.0.1:8000/admin/login
Chat
FitBuddy AI
Generator
Admin
API Docs
ADMIN
Sign in
Username
Password
Sign in
FitBuddy is an educational wellness-planning tool, not medical advice.


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WhatsApp
FitBuddy – AI Fitness Plan Gene
FitBuddy – AI Fitness Plan Gene
Admin Dashboard — FitBuddy
127.0.0.1:8000/view-all-users
Chat
FitBuddy AI
Generator
Admin
API Docs
ADMIN DASHBOARD
Log out
Users and workout plans
Siva (user001)
Delete
Age 25 · 65.0 kg · weight loss · medium intensity
Original plan
Updated plan
No feedback update yet.
FitBuddy 7-Day Plan for Siva
Goal: weight loss | Intensity: medium
Training emphasis: cardio + full-body conditioning
Safety note: use a comfortable range of motion and stop if you feel pain, dizziness, or unusual symptoms.
Day 1
Focus: Full Body
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Squats, push-ups, rows. Suggested volume: 3 sets × 8–12 reps.
Rest: 60–90 seconds between strength sets, as appropriate.
Cooldown: 5–10 minutes of easy movement and gentle stretching.


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WhatsApp
FitBuddy – AI Fitness Plan Gene
FitBuddy – AI Fitness Plan Gene
FitBuddy API - Swagger UI
127.0.0.1:8000/docs
Chat
FitBuddy API
1.0.0
OAS 3.1
/openapi.json
AI fitness plan generator based on the supplied FitBuddy project specification.
default
GET / Home
POST /generate-workout Generate Workout
POST /submit-feedback Submit Feedback
GET /view-all-users View All Users
POST /admin/delete/{user_id} Admin Delete User
GET /admin/login Admin Login Page
POST /admin/login Admin Login
POST /admin/logout Admin Logout
POST /api/generate-workout Api Generate Workout
POST /api/submit-feedback Api Submit Feedback
GET /api/users Api Users
GET /api/users/{user_id} Api User
GET /health Health